Response to Reviewer 2

Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing

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Dear Reviewer 2,

We thank you for your review and comments, which have helped us to improve the manuscript considerably. Below we provide a point-by-point response to your comments, and we have revised the manuscript accordingly.

Comments 1: *The abstract is clear and presents the manuscript; however, it is too long (235 words). It must be less than 200 words.*

Response 1: We rewrote the abstract, which is now shorter than 200 words.

Revised in: Abstract (page 1, lines 2–17).

Comments 2: *Some of the keywords are wrong, such as "policy privacy" – 4 times in the text and "data use ontology" – 7 times.*

Response 2: We revised the keywords to match the terms used in the manuscript. "Policy privacy" is now "consent policy privacy", the phrase in the title, which the Introduction and Section 4.3 now also use for the property we evaluate; "data use ontology" is now "Data Use Ontology (DUO)", whose abbreviation the text uses throughout; and "zero-knowledge proof" is now "zero-knowledge proofs". We also added "metadata leakage", the central concern of the paper, which the Introduction and Related Work now name explicitly.

Revised in: Keywords (page 1, lines 18–19); Section 1, third paragraph (page 2, lines 44–50); Section 4.3, first paragraph (page 10, lines 366–370).

Comments 3: *The introduction correctly outlines the issues. However, rows 123–131 point to old sources and methods of encryption. It raises the question "why?" – do the authors not know the newest or do they not want to use the newest; why do the authors not comment on the newest solutions?*

Response 3: We kept the CP-ABE and PRE citations because they define the primitives this paragraph describes. We also added recent blockchain healthcare designs built on them: PRE-based sharing of medical records and IoMT data [19,20] and policy-hiding CP-ABE, one of them with distributed key generation [13,14]. The Related Work section also discusses recent hidden-policy, self-sovereign-identity, and zero-knowledge authentication work (see Response 10).

Revised in: Section 2.1 (pages 3–4, lines 122–135); Section 3, third and fourth paragraphs (pages 5–6, lines 190–210; page 6, lines 211–233).

Comments 4: *There is so much empty space on page 6, page 20, and page 25 because of the wrong position of Table 2, Table 9, and Table 12.*

Response 4: All tables and figures now use standard float placement, we adjusted the float-page settings, and the abbreviations list can now break across pages.

Revised in: Whole manuscript; the former Tables 2, 9, and 12 are now Tables 2 (page 7), 11 (page 26), and 14 (page 29).

Comments 5: *The titles of tables and figures are very long. For example, Figure 1 has a 4-row title and Table 12 a 6-row title.*

Response 5: We updated every caption to one or two sentences and moved the explanations into the main text.

Revised in: All figure and table captions, for example Figure 1 (page 12) and Table 14 (page 29).

Comments 6: *On row 681 there is a formula in the text. This is not acceptable. Follow the proper format of formulas.*

Response 6: We removed the inline formulas from the results text. Statistics are now given as plain numbers (for example, "mean AUC 0.667 with 53% of seeds significant"), and the full values remain in the tables. The formulas that remain are displayed and numbered: the permutation p-value and the AUC gap in Section 6.2.1 (Equations (16) and (17)), the gas decomposition in Section 6.4 (Equation (18)), and the paired leakage metric in Section 6.9 (Equation (19)).

Revised in: Section 6.2.1, paragraphs on the permutation test and the AUC gap (page 19, lines 680–693); Section 6.4, paragraph "Supervised attack" and the paragraph that follows (page 23, lines 837–848) and paragraph "What the ch5 classifier separates" (pages 23–24, lines 849–878); Section 6.5, last paragraph (pages 24–25, lines 900–910); Section 6.6, paragraph "Supervised attack" and the paragraph that follows (pages 25–26, lines 927–947); Section 6.9, fourth paragraph (page 29, lines 1023–1031) and paragraph introducing Figure 3 (page 29, lines 1036–1043).

Comments 7: *The opinion of the authors about the results is very short. The conclusions in Part 6 are almost missing.*

Response 7: We have expanded both parts. The Discussion now opens with our interpretation of the results, including the two criterion failures and why we report them as failures. The Conclusion has been rewritten to state the contribution, the main findings, and the scope within which they hold.

Revised in: Section 7, paragraphs under "Privacy findings" (pages 31–33, lines 1122–1145); Section 8 (pages 34–35, lines 1229–1244).

Comments 8: *The English language looks quite good. The title and abstract reflect the content.*

Response 8: Thank you for the positive feedback.

Comments 9: *The discussion and conclusion summarize the proposed idea. However, there is a repetition of the same conclusion in the abstract, discussion, and conclusion. This repetition of the same information should be avoided.*

Response 9: We removed the repetition. The Abstract states the results and overhead figures once. The Discussion interprets them (the two criterion failures, cost, and trust model) without restating the Abstract. The Conclusion summarizes the contribution, the findings, and the assumption under which they hold, without repeating the task counts or overhead figures.

Revised in: Abstract (page 1, lines 2–17); Section 7, paragraphs under "Privacy findings" (pages 31–33, lines 1122–1145) and "The cost of padding" (page 33, lines 1146–1168); Section 8 (pages 34–35, lines 1229–1244).

Comments 10: *Many of the references are old [7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24,*

25, 26, 27, 30, 31]. Some of them do not follow the proper format. I offer the authors to find newer sources and replace these old sources.

Response 10: We added seven references from 2024–2026 and cite them where the text relies on them. They cover PRE-based medical data sharing [19,20], policy-hiding CP-ABE [13,14], self-sovereign identity [27], zero-knowledge patient authentication [28], and a survey of zero-knowledge proofs for blockchain privacy [29]. The list now has 41 entries, 12 of them from 2024 or later (previously 31 and 5); the total also includes three foundational references for the Poseidon hash, EdDSA, and the Baby Jubjub curve, which the revised trust model now introduces explicitly. We kept the older references that are original constructions or standards used in the paper (for example Groth16, Zerocash, CP-ABE, PRE, ICD-10, and OBI), since no newer source replaces them. We also corrected the entry types and formats of several references, including those of Kakarlapudi and Mahmoud, AZTEC, Zeth, and Ibraimi et al., and added DOIs.

Revised in: Section 2.1 (pages 3–4, lines 122–135); Section 3, third and fourth paragraphs (pages 5–6, lines 190–210; page 6, lines 211–233); References.